

किताबें किसी के लिए सोती नहीं,
ये उनके लिए जागती रहती हैं,
जो जागते रहते हैं
किताबों के लिए।



DATA BASE SQL

WHY WE NEED DATABASE

Aniket → No., Name, dept, Year

file system: detail.txt

					Marks
①	Aniket	=	—	—	—
②	Sushant	=	—	—	—
③	Snehal	=	—	—	50 90
④	Pranali	=	—	—	—

1000

- duplicate data
- Difficulty accessing data
- No security for data
- Difficulty - Search, delete, update
- Too many users can access the file and they can R-W-U

WHAT IS DATABASE

It is a collection of inter-related data.



* DBMS (Database Mgm. System)

- set instⁿ / prog. to perform
optⁿ CRUD on DB.

-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
DB			

- ① What is DB
- ② What is DBMS
- ③ Types of DBMS

* Types of Database

— Relation : to store data in Tabular format
(Rows & column)

— Non-Relation $\Rightarrow$ ① XML
② JSON

Roll	Name	Mark	Age
101	Kush	98	10
102	Raj	93	12
103	Ravi	98	10
104	Kush	94	20
105	Raj	92	12

Relation

JSON

{

"name" : "Ravi",

"roll" : 103,

"marks" : 98,

"Age" : 12

}

Non-Relation

column / Property / Attribute

Roll	Name	Mark	Age
101	Kush	98	10
102	Raj	93	12
103	Ravi	98	10
104	Kush	94	20
105	Raj	92	12

Row / Tuple / Record

FUNCTIONAL DEPENDENCY

A dependency $A \rightarrow B$ is called F.D if whenever we are getting a unique value of A it always gives unique value on B

✗ Name $\rightarrow$ Marks FD?

Marks $\rightarrow$ Age ✓

————— ✗ —————

① Roll $\rightarrow$ Age ✓

② Name $\rightarrow$ Age ✗

③ Age $\rightarrow$ Roll ✗

Roll	Name	Mark	Age
101	Kush	98	10
102	Raj	93	10
103	Ravi	98	10
104	Kush	94	10
105	Raj	92	10

(Left)

(Right)

A $\longrightarrow$ B

(

if all are

Unique, then

100% FD

(

if all are same

then 100% FD

TYPES OF FUNCTIONAL DEPENDENCY

TRIVIAL FUNCTIONAL DEPENDENCY

$A \rightarrow B$ is called Trivial FD iff B is a subset of A

$$AB \rightarrow B,$$

Non - TRIVIAL FUNCTIONAL DEPENDENCY

$A \rightarrow B$ is called Non-Trivial iff

$$A \cap B = \emptyset$$

$ABC \rightarrow D$

$ABC \rightarrow CD$

N.TX

TX

Semi-Trivial
or

Semi Non-Trivial

$A \rightarrow B \rightarrow NT$

$AB \rightarrow C \rightarrow NT$

$ABC \rightarrow AC \rightarrow \text{Trivial}$

$A \rightarrow BCD \rightarrow NT$

$ABC \rightarrow DEF \rightarrow NT$

$A \rightarrow A \rightarrow \text{Trivial}$

$\phi \rightarrow \phi \rightarrow \underline{\text{Trivial}}$

$AB \rightarrow D \rightarrow NT$

$ABC = \{ \phi, A, B, C, AB, BC, AC, ABC \}$

$CD = \{ \phi, C, D, CD \}$

KEYS

KEYS

It is a non-empty set of attributes with unique values.

CANDIDATE KEYS

A set of one or more columns that can uniquely identify a row in table

PRIMARY KEYS

Any Candidate key can be selected as primary key.

* Closure Attributes

$R (A B C D E)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, BC \rightarrow E \}$

$A^* \rightarrow \{ A B C D E \}$

$B^* \rightarrow \{ B C D E \}$

$C^* \rightarrow \{ C D \}$

$D^* \rightarrow \{ D \}$

$E^* \rightarrow \{ E \}$

HOW TO FIND CANDIDATE KEYS

$R(A B C D)$ Attribute
|
Relation $\rightarrow$ Table

$R(A B C D)$

FD = $\{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

* Step 1: cut right side Attributes

* Step 2: Write down Remaining Attribute

* Step 3: find closure of A & check if it contains all the Reln Attribute

① if yes $\rightarrow$ CK

② if NO $\rightarrow$ try with combination

$R(A B C D)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

RHS: BCD :

LHS: A

$A^* = \{ A B C D \}$



$R(ABCDE)$

FD: $\{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E \}$

RHS: BCDE

LHS: A

$A^* = \{ ABCDE \}$



R (A B C D E)

FD : $\{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A \}$

RHS: B C D E A

LHS: None

CK

$A^+ = \{ A B C D E \}$

$D = \{ D E A B C \}$

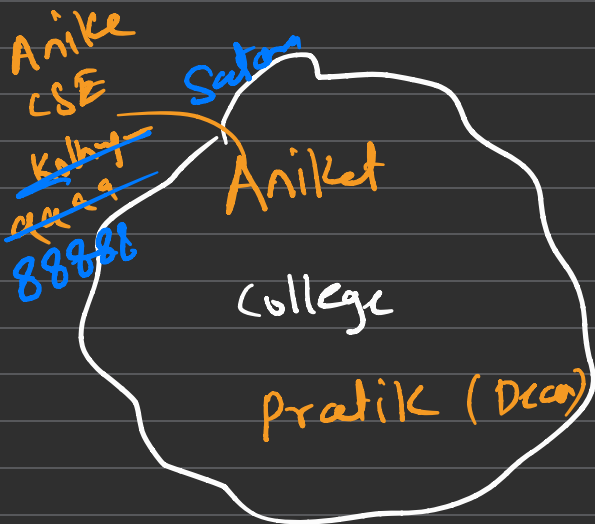
$E^+ = \{ E A B C D \}$

$C = \{ C D E A B \}$

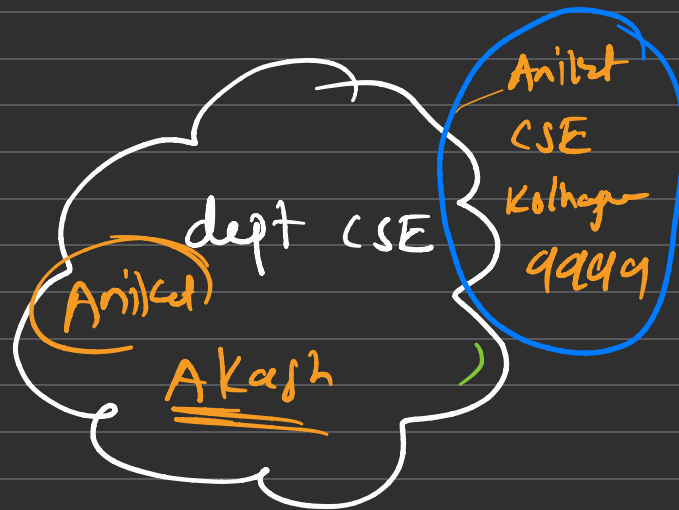
$B^+ = \{ B C D E A \}$

WHY WE NEED NORMALIZATION

- To Remove Data Redundancy
- Convert large Relation to small Relation



- INSERT
- Delete
- Update
- Redundancy



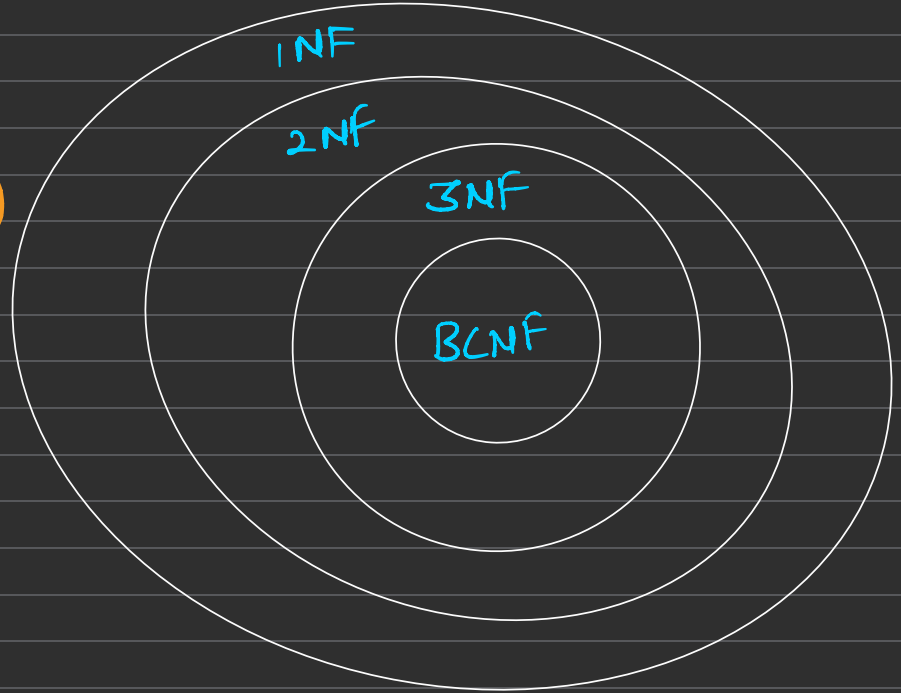
WHAT IS NORMALZIATION

$BCNF \subseteq (1NF \ 2NF \ 3NF)$

$3NF \subseteq (1NF \ 2NF)$

$2NF \subseteq (1NF)$

$1NF \rightarrow$ By default



Attributes

- Single Valued:

Attribute which has single value

Eg: Adhar card, Pan card, Voting card

- Composite Attribute

Attribute which further get divided

Eg: full Name → FN MN SN

- Multi-Valued

Attribute which has multiple value

Eg: phone No., Email

- Derived Attribute :

Attribute which can be derived from other

Attribute DOB → Age

TYPES OF NORMALIZATION

1 NF

2 NF

3 NF

BCNF

1 NF

Relation is said to be in 1NF iff

- ① It does not contain composite or multi-valued attribute

ID	Name	Email
1	Raj	x, y
2	Ravi	a
3	Yash	b
4	Kush	z, m

Convert
→
1NF

ID	Name	Email
1	Raj	x
1	Raj	y
2	Ravi	a
3	Yash	b
4	Kush	z
4	Kush	m

* So in 1NF we are unable to Remove anomalies ,

Hence we need 2NF

2 NF

Relation is said to be in 2NF iff

- ① It must be in 1NF
- ② There should not be 'partial functional dependency'

OR

- ③ Non-prime Attribute should be fully dependent on PK/CK

* Partial FD :

Non-prime Attribute depends on Prime Attribute

(Prime) A $\rightarrow$ B (Non-Prime)

$R(ABCDE)$

$FD = \{ A \rightarrow B, B \rightarrow C, D \rightarrow C, D \rightarrow E \}$

RHS: BCE

(P) $A \rightarrow B$ (NP)

LHS: AD

$AD^* = \{ AD BCE \} = \underline{\underline{CK}}$

- prime Attribute = part of CK = A, D

- Non-prime Attribute = non-part of CK = BCE

$A \rightarrow B, D \rightarrow C, D \rightarrow E \Rightarrow$ Partial FD

Not in 2NF

$R(ABCD)$

$FD = \{ A \rightarrow B \quad B \rightarrow C \quad C \rightarrow D \quad D \rightarrow A \}$

$RHS = B C D A$

prime Attributes = NO

$LHS = -$

Non-prime = NO

$A^* = \{ ABCD \}$

$CK = A, D, C, B$

2NF

$R(ABCDE)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E \}$

$RHS = BCDE$

$LHS = A$

$A^* = \{ ABCDE \} = \underline{\underline{CK}}$

2NF

prime Attribute = —

Non-prime Attribute = B, C, D, E

3 NF

Relation is in 3NF iff

- ① it must be in 2NF
- ② There should not be 'Transitive Dependency'

* Transitive dependency

Non-prime should not depend on Non prime Attribute

~~Non-prime $\longrightarrow$ Non-Prime~~

- Step ① : find CK
- Step ② : find Prime & Non-prime Attribute
- Step ③ : Check if it is 2NF
- Step ④ : Check if it is 3NF

R (ABCDE)

FD = { $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$ }

RHS: BCDE

LHS: A

It is in 2NF

$A^* = \{ABCDE\} = \text{CK}$

① prime Attribute = Null

② Non-prime = B, C, D, E

Not in
3NF

$R(ABCD)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A \}$

RHS = BCDA

LHS = —

$A^+ = \{ ABCD \}$

CK = A, B, C, D

prime = Null

Non-prime = Null

3NF

$R(ABCDE)$

$FD = \{ A \rightarrow BC \quad B \rightarrow DE \quad DE \rightarrow A \}$

$RHS = BCDEA$

$LHS = \text{NULL}$

— 2 NF

$A^* = \{ ABCDE \}, DE, B$

prime Attribute = D, E

Non-prime = C

3 NF

BCNF

Relation is in BCNF iff

①. must be in 3NF

②. $X \twoheadrightarrow Y \Rightarrow X$ must be CK

$R(ABCDE)$

$FD = \{ A \rightarrow B, \underbrace{B \rightarrow C}, \underbrace{C \rightarrow D}, \underbrace{D \rightarrow E} \}$

Transitive
dependency

$RHS = BCDE$

prime = Null

$LHS = A$

Non-prime = B, C, D, E

$A^* = \{ ABCDE \}$

- 2NF ✓

- 3NF ✗

- BCNF ✗

$R(ABCD)$

$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow A, D \rightarrow A \}$

$RHS = BCA$

$LHS = D$

$D^* = \{ DABCD \}$

prime = Null

Non-Prime = A B C

- 2NF ✓

- 3NF ✗

- 4NF ✗

R (ABCDE)

FD: $\{ A \rightarrow BC, DE \rightarrow A, C \rightarrow DE \}$

RHS = BC A DE :

prime = D, E

LHS = —

Non-prime B

$A^* = \{ ABCDE \}, DE, C$

☒ 2NF

☒ 3NF

☒ BCNF

ACID PROPERTY



Relational DB

✓ Atomicity

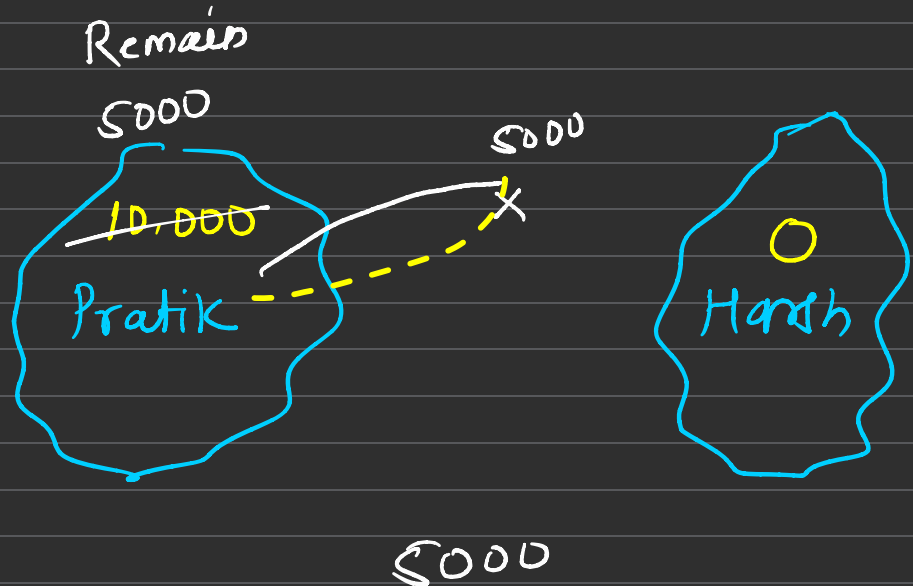
✓ Consistency

✓ Isolation

✓ Durability

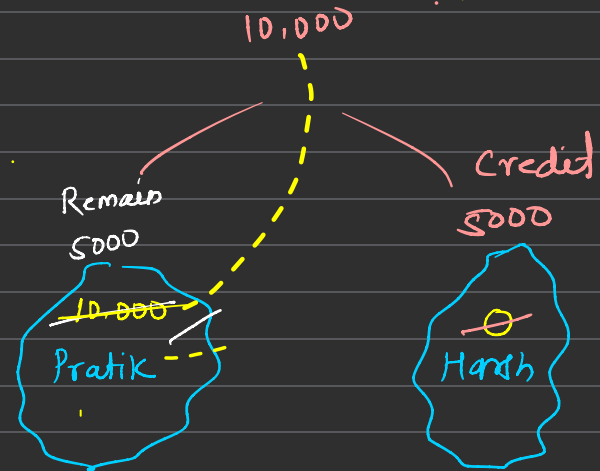
Atomicity:

either perform all transaction or
don't perform



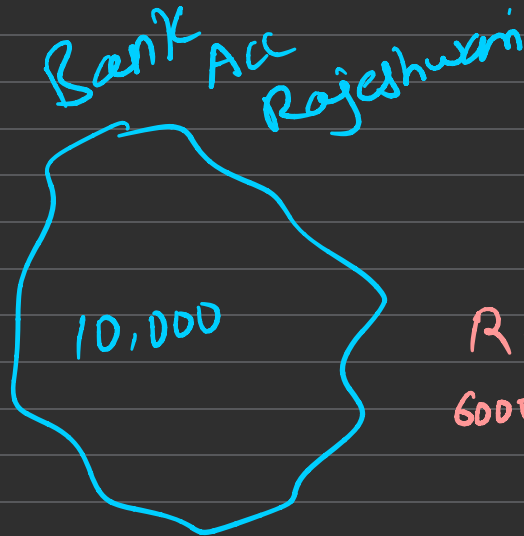
Consistency

data should remain consistent before and after performing transaction.



Isolation

Rajeshwari ^① → 2 ATM Cards
Snehal — ①

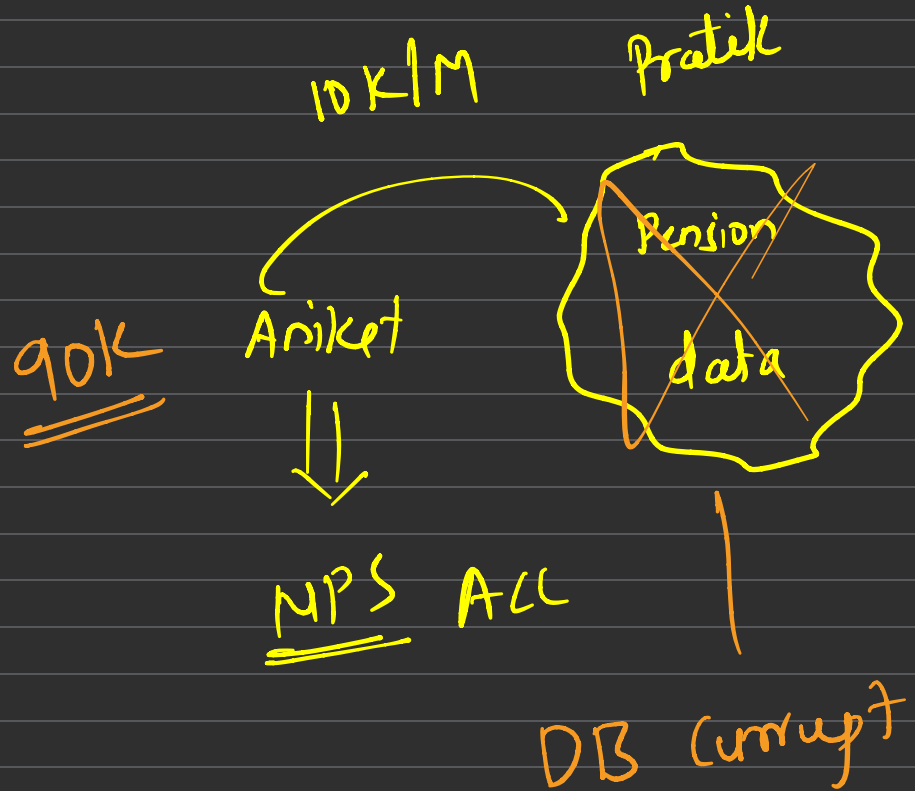


4000 Withdraw at a time

ATM

ATM 3000 withdraw

Durrability



Server-US

insta, yt,
whatsapp

CDN = Content Delivery Network

